

JAYPEE UNIVERSITY OF ENGINEERING & TECHNOLOGY, GUNA
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Course: Computer Organization & Architecture Lab

Course Code: CS208

B. Tech. (CSE IV Sem.)

Experiment # 7

Aim: Design of Arithmetic Logic Unit (ALU) and Multiplier.

An arithmetic logic shift unit (generally known as ALU) is one of the important units of the central processing unit (CPU) of a computer system. It does all processes related to arithmetic and logic operations that need to be done on instruction words. In some microprocessor architectures, the ALU is divided into the arithmetic, logic and shift units. The design and implementation of the ALU is an important architecture design problem. Block diagrams of four operations ALU with two control (select) lines are shown here below:-

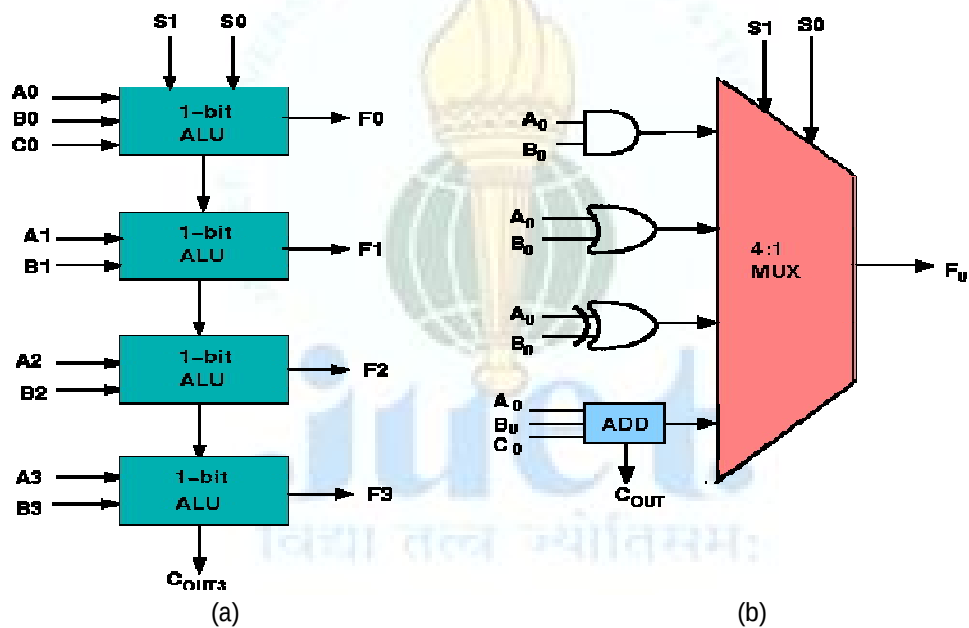


Fig. 1: Block diagrams (a) 4-bit-4-operation ALU (b) 1-bit-4-operation ALU

Exercise#1: Design 4-bit, 8-operations (given in Table) Arithmetic Logic Unit (ALU) using initial carry (C_{in}) as LSB of the control lines of multiplexers as shown in block diagram given below. Use control signals and input/output signals in the form of **bus**. Show logic diagrams Arithmetic unit, logic unit, shift unit and complete block diagram ALU in your file.

Control Signals	Operations
$S_2S_1S_0$	Arithmetic Operations
000	Transfer : $F = A$
001	Subtract: $F = A - B$
010	Addition: $F = A + B$
011	Decrement: $F = A - 5$
Logical Operations	
100	XOR: $F = A \oplus B$
101	AND: $F = A \wedge B$
Shift Operations	
110	Circular left shift of A: $F = cil A$
111	Arithmetic shift of B: $F = Ashr B$

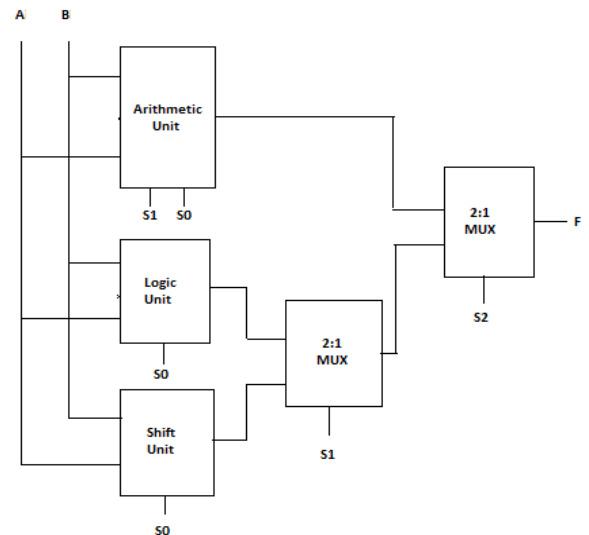


Figure-2: Block diagram of 4-bit, 8-operation ALU

Design steps:

- Use **4 bit bus in-built blocks** of multiplexers, shifters, full adder, gates required in the design.
- Design arithmetic, logic and shift units individually and use them as **sub-circuit** in the design of ALU using input/output signals in the form of **bus**.
- Implement the design of ALU.
- Add two hex displays at input and two hex displays at output (one with 4bit output and one with one bit end carry).
- To verify the ALU design, apply four 4-bit random inputs, vary the selection lines, and check the output if it is correct or not.

Exercise#2: Design 2-bit array multiplier using **in-built block** of half adder. Display both the inputs and output result in decimal format using hex display devices.